AI/ML intern DAY 6

TASK 5 — INTEGRATING AI OUTPUTS WITH BACKEND APIs

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1. Abstract

This task focused on integrating Artificial Intelligence outputs with backend API systems using Python and Flask.

The project explored how AI-generated outputs such as:

- pose estimation results
- body measurements
- recommendation results
- segmentation data
- AI predictions

can be transferred to backend systems using APIs for real-time applications.

The task also studied how AI systems communicate with web applications, databases, and frontend interfaces through backend APIs.

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2. Objective

The objective of this task was to understand how AI systems connect with backend APIs and transfer AI-generated outputs dynamically.

The task focused on:

- AI and backend integration
- API communication
- JSON response handling
- AI output transfer
- real-time AI systems
- backend workflow management

3. Introduction to AI Backend Integration

Modern AI systems do not work independently.
Most AI applications send their outputs to backend servers using APIs.

The backend API acts as a communication layer between:

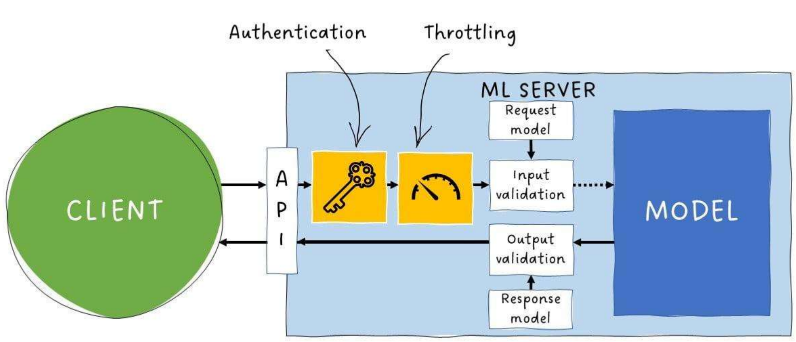
- AI models
- frontend applications
- databases
- web systems
- mobile applications

The AI system generates outputs such as:

- body measurements
- recommendation results
- segmentation masks
- pose detection data

These outputs are transferred using:

- JSON responses
- REST APIs
- HTTP requests



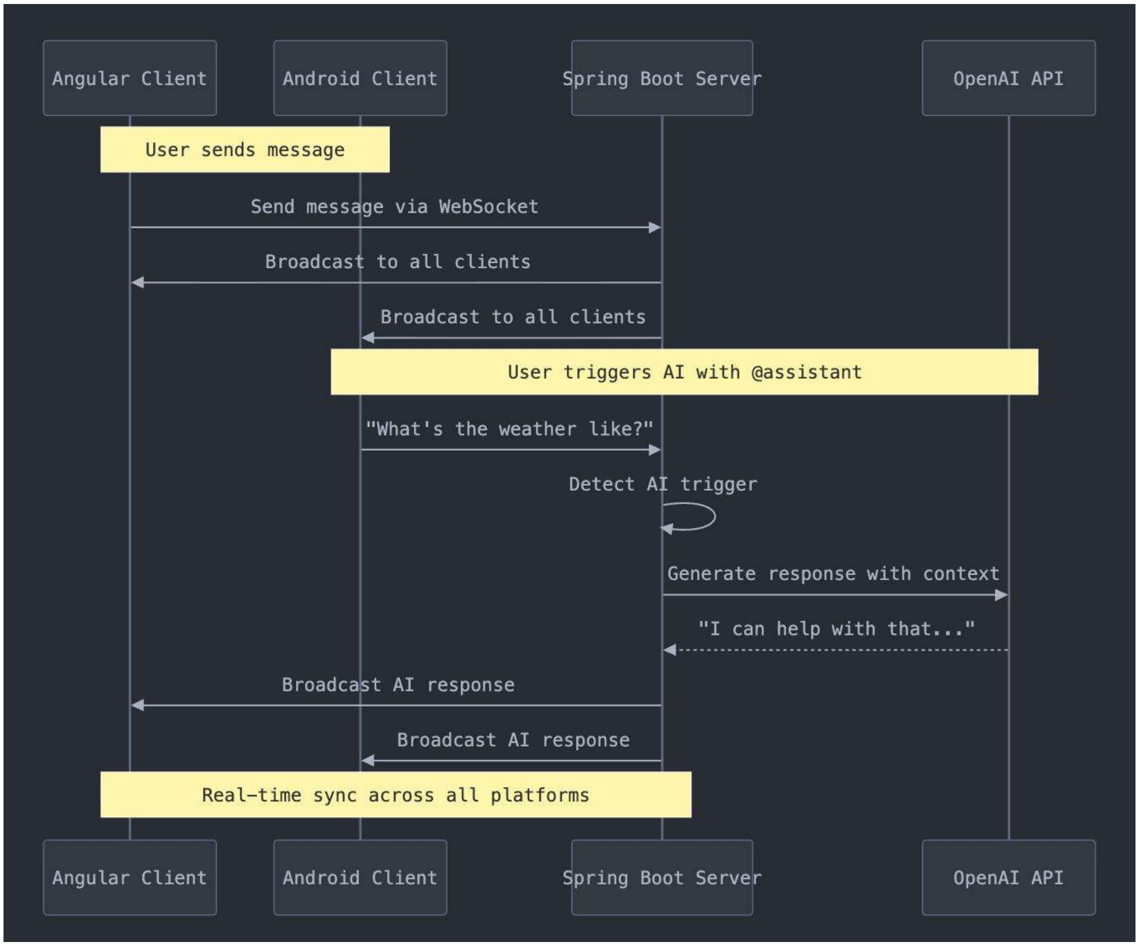


Figure 1 — AI Backend API Communication

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4. Workflow of AI API Integration

The backend integration workflow consists of multiple stages.

Stage 1 — Input Collection

The AI system receives:

- images
- webcam frames
- user preferences
- body landmark data

Stage 2 — AI Processing

The AI model performs:

- pose estimation
- segmentation
- recommendation generation
- body analysis

Stage 3 — Output Generation

The AI system generates:

- prediction results
- recommendation outputs
- body measurements
- segmentation information

Stage 4 — API Communication

The backend API transfers:

- AI outputs
- JSON data
- prediction results
- system responses

to frontend systems.

Stage 5 — Final Visualization

The frontend application displays:

- AI predictions
- outfit recommendations
- body tracking results
- virtual fitting outputs

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5. Technologies Used

Technology	Purpose
Artificial Intelligence	Prediction generation
Flask API	Backend communication
Python	AI implementation
JSON	Data transfer
REST API	API communication
Computer Vision	AI image analysis

6. API Response Structure

AI systems usually transfer outputs using JSON format.

Example JSON Response

```
{
  "body_measurement": "42 cm",
  "pose_status": "Detected",
  "recommended_outfit": "Hoodie + Jeans",
  "confidence": 0.94
}
```

7. Pseudo Code — AI API Integration Workflow

START

Capture input data

Process input using AI model

Generate AI prediction

Convert output into JSON format

Send response through backend API

Display final output on frontend

END

8. Algorithm — AI Backend Integration System

Step-by-Step Algorithm

Step 1

Initialize Flask backend server.

Step 2

Load AI prediction model.

Step 3

Capture user input or image data.

Step 4

Process input using AI system.

Step 5

Generate AI prediction output.

Step 6

Convert prediction into JSON response.

Step 7

Send output through API endpoint.

Step 8

Display final response on frontend application.

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9. Python Backend API Implementation

Flask API Example

```
from flask import Flask, jsonify
app = Flask(__name__)
@app.route('/ai-output')
def ai_output():
    result = {
        "pose_detection": "Successful",
```

```
        "body_measurement": "42 cm",
        "recommended_outfit": "Hoodie + Jeans",
        "confidence": 0.94
    }
    return jsonify(result)

if __name__ == '__main__':
    app.run(debug=True)
```

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10. How the API Works

The API workflow functions as follows:

Step	Process
1	User sends request
2	Backend receives request
3	AI model processes data
4	AI generates prediction
5	Backend converts output into JSON
6	Frontend displays AI output

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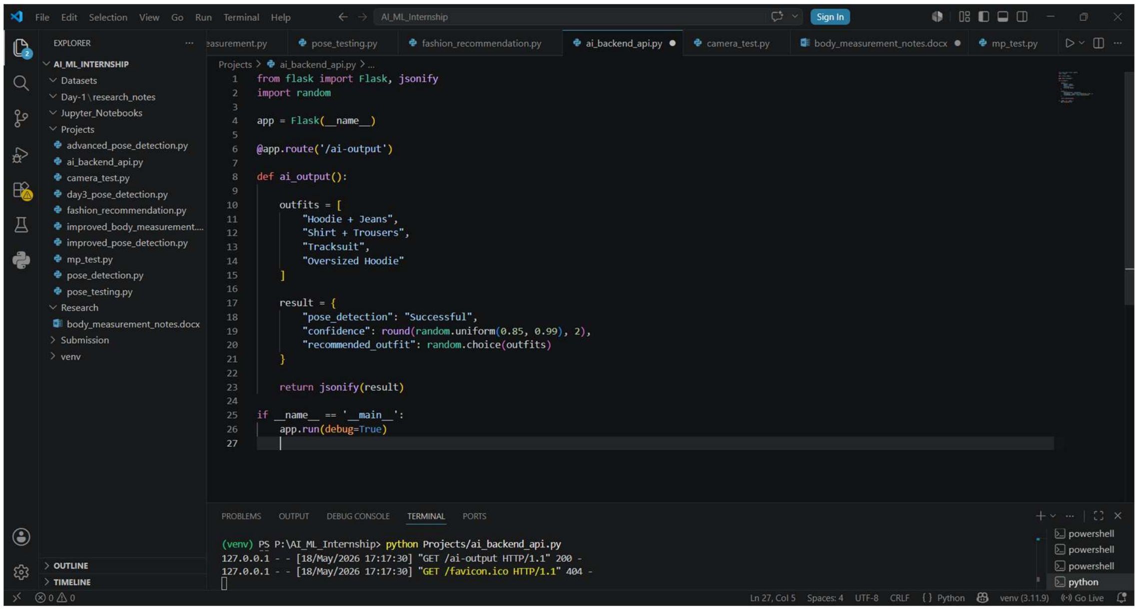
11. Running the Backend API

Install Flask

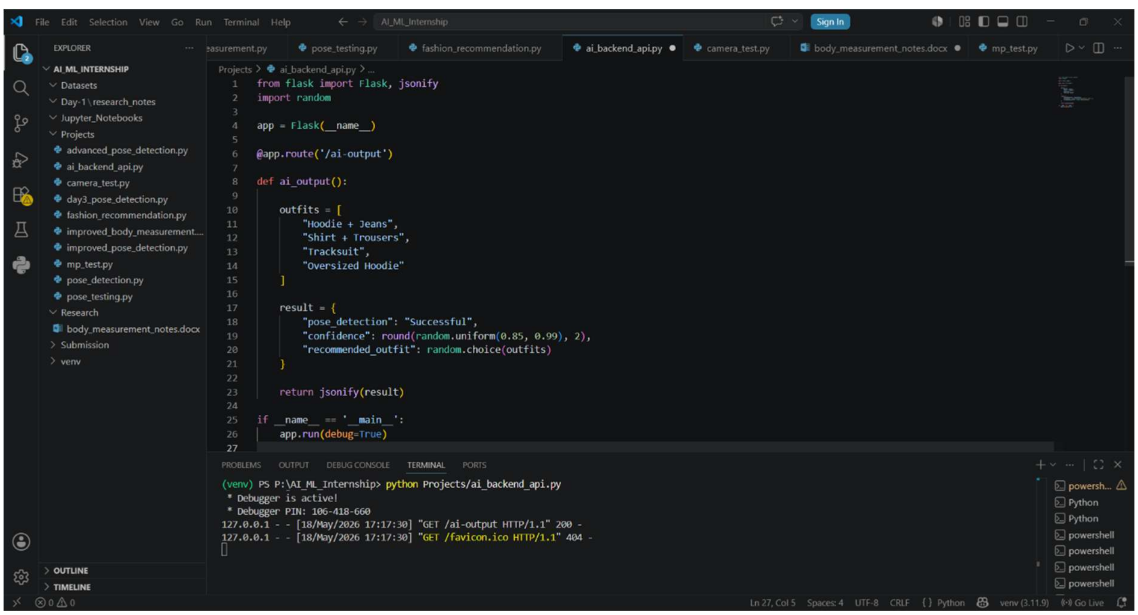
Run:

pip install flask



Run the API

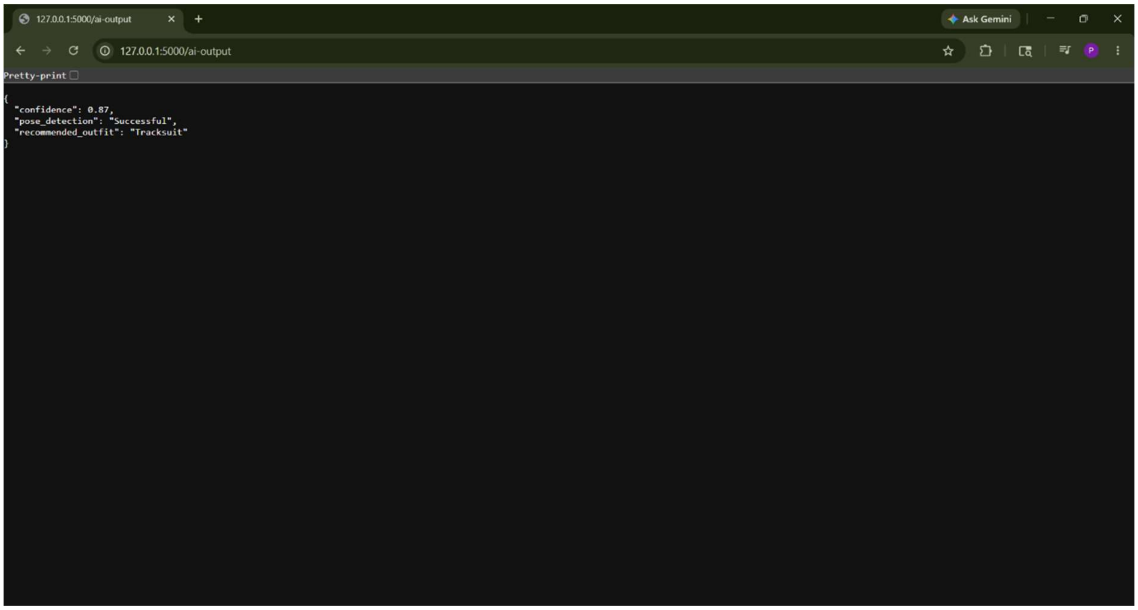
python Projects/ai_backend_api.py



Open Browser

Open:

http://127.0.0.1:5000/ai-output



The browser displays AI-generated JSON output.

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12. Applications of AI Backend APIs

AI backend APIs are used in:

- AI fashion systems
- recommendation systems
- virtual try-on applications
- healthcare AI systems
- AI assistants
- real-time computer vision systems
- smart shopping platforms

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13. Challenges Faced

Challenge	Impact
Slow API response	Delayed predictions
Large AI models	High processing time
JSON formatting errors	Incorrect responses
Real-time processing	Increased system load

14. Knowledge and Skills Acquired

During this task, the following concepts were understood:

- AI backend integration
- Flask API development
- JSON response handling
- AI output communication
- REST API workflows
- backend AI systems
- frontend-backend interaction

15. Conclusion

Integrating AI outputs with backend APIs is an important part of modern Artificial Intelligence systems.

The task improved understanding of:

- AI backend communication
- API workflows
- AI output transfer
- JSON response systems
- Flask API development
- real-time AI integration

These technologies are widely used in intelligent fashion systems, recommendation engines, virtual try-on platforms, and modern AI web applications.